MSMS 408 : Practical 16

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Question

Consider a random variable $X \sim \mathscr{B}e(2,5)$ with mean $E[X] = \frac{2}{7}$.

(i) Estimate the mean using Monte Carlo simulation and the antithetic variable technique. Compare the variances of the two techniques.

(ii) Estimate $E[e^{-X}]$ using Monte Carlo simulation and the antithetic variable technique. Compare the variances of the two techniques.

(iii) Estimate $E[X^2]$ using Monte Carlo simulation and the antithetic variable technique. Compare the variances of the two techniques.

Theory

- To generate random numbers from $\mathscr{B}e_1$ distribution, we recall

$$X \sim U(0,1) \Rightarrow -\ln X \sim \text{ Exp}(1).$$

  Also,

$$\sum_{i=1}^{n} -\ln X_i \sim \text{ Gamma(shape} = n, \text{scale} = 1) \qquad \text{and}$$

$$\frac{S_1}{S_1 + S_2} \sim \beta_1(m,n)$$

  where S_1 and S_2 are independent Gamma distributions with shape parameters m and n respectively.

- If $x_1, x_2, \ldots, x_n$ is a sample of size n from X, Monte Carlo estimate of $\theta = E[X]$ is

$$\widehat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

  Estimate of θ by using antithetic variable will be

$$\widehat{\theta} = \frac{1}{n}\sum_{i=1}^{n/2}[X_i + X_i']$$

  where X_i and X_i' are negatively correlated random variables.

- If $x_1, x_2, \ldots, x_n$ is a sample of size n from X, Monte Carlo estimate of $\theta = E[e^{-X}]$ is

$$\widehat{\theta} = \frac{1}{n}\sum_{i=1}^{n} e^{-x_i}.$$

Estimate of θ by using antithetic variable will be

$$\widehat{\theta} = \frac{1}{n}\sum_{i=1}^{n/2}[e^{-X_i} + e^{-X_i'}]$$

where X_i and X_i' are negatively correlated random variables; $g(x) = e^{-x}$ is a monotone decreasing function of x, consequently e^{-X_i} and $e^{-X_i'}$ are negatively correlated random variables.

- If $x_1, x_2, \ldots, x_n$ is a sample of size n from X, Monte Carlo estimate of $\theta = E[X^2]$ is

$$\widehat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i^2.$$

Estimate of θ by using antithetic variable will be

$$\widehat{\theta} = \frac{1}{n}\sum_{i=1}^{n/2}[X_i^2 + X_i'^2]$$

where X_i and X_i' are negatively correlated random variables; $g(x) = x^2$ is a monotone increasing function of x, consequently X_i^2 and $X_i'^2$ are negativly correlated random variables.

- Here

$$X_i = h(\underset{\sim}{u}), \qquad \& \qquad X_i' = h(\mathbf{1} - \underset{\sim}{u})$$

with

$$\underset{\sim}{u} = (u_1, u_2, \ldots, u_7), \qquad \text{and}$$

$$h(u_1, u_2, \ldots, u_7) = \frac{\sum_{i=1}^{2} - \ln u_i}{\sum_{i=1}^{2} - \ln u_i + \sum_{i=3}^{7} - \ln u_i}.$$

➜ R Program

```
h <- function(u, m, n){
  u1 <- u[1:m]
  u2 <- u[(m + 1) : (m + n)]

  x1 <- -sum(log(u1))
  x2 <- -sum(log(u2))

  x <- x1 / (x1 + x2)
}
```

```r
my.rbeta <- function(n, shape1, shape2, uniform.sample){

  s <- c()

  for (i in 1:n) {
    s[i] <- h(uniform.sample[i,], shape1, shape2)
  }

  return(s)
}
```

```r
sample.sizes <- c(20, 50, 80, 100)
```

1

```r
theta.hat.monte.carlo <- c()

for (i in 1:length(sample.sizes)) {

  size <- sample.sizes[i]

  u <- matrix(runif(size * (2 + 5)), nrow = size, ncol = (2 + 5))

  theta.hat.monte.carlo[i] <- mean(my.rbeta(size, 2, 5, u))
}
```

```r
theta.hat.antithetic <- c()

for (i in 1:length(sample.sizes)) {
  size <- sample.sizes[i]

  u <- matrix(runif((size / 2) * (2 + 5)), nrow = size / 2, ncol = (2 + 5))

  x <- my.rbeta(size / 2, 2, 5, u)
  x_dash <- my.rbeta(size / 2, 2, 5, 1-u)

  theta.hat.antithetic[i] <- mean(c(x, x_dash))
}
```

```r
df1 <- data.frame(sample.size = sample.sizes,
                  theta.hat.mc = theta.hat.monte.carlo,
                  theta.hat.at = theta.hat.antithetic)
```

Table 1: Estimates from Two Different Techniques

sample.size	theta.hat.mc	theta.hat.at
20	0.199437	0.289805
50	0.287849	0.286490
80	0.291264	0.290699
100	0.283530	0.293247

True value is $\theta = E[X] = \dfrac{2}{7} = 0.2857143$.

```r
B <- 1000
```

```r
var.monte.carlo <- c()

for (j in 1:length(sample.sizes)) {

  size <- sample.sizes[j]

  theta.hat.monte.carlo <- c()

  for (i in 1:B) {
    u <- matrix(runif(size * (2 + 5)), nrow = size, ncol = (2 + 5))

    theta.hat.monte.carlo[i] <- mean(my.rbeta(size, 2, 5, u))
  }

  var.monte.carlo[j] <- var(theta.hat.monte.carlo)
}
```

```r
var.antithetic <- c()

for (j in 1:length(sample.sizes)) {

  size <- sample.sizes[j]

  theta.hat.antithetic <- c()

  for (i in 1:B) {
    u <- matrix(runif((size / 2) * (2 + 5)), nrow = size / 2, ncol = (2 + 5))

    x <- my.rbeta(size / 2, 2, 5, u)
    x_dash <- my.rbeta(size / 2, 2, 5, 1-u)

    theta.hat.antithetic[i] <- mean(c(x, x_dash))
  }

  var.antithetic[j] <- var(theta.hat.antithetic)
}
```

```r
relative.gain <- ((var.monte.carlo - var.antithetic) / var.monte.carlo) * 100
```

```r
df2 <- data.frame(sample.size = sample.sizes,
                  variance.mc = var.monte.carlo,
                  variance.at = var.antithetic,
                  percentage.reduction = relative.gain)
```

Table 2: Comparison of Variance of Two Techniques

sample.size	variance.mc	variance.at	percentage.reduction
20	0.001220	0.000336	72.428670
50	0.000507	0.000128	74.805150
80	0.000331	0.000081	75.702100
100	0.000252	0.000063	74.983230

✎ We achieve significant reduction in variance of the estimates by using antithetic variable technique.

2

```
g <- function(x) exp(-x)
```

```
theta.hat.monte.carlo <- c()

for (i in 1:length(sample.sizes)) {

  size <- sample.sizes[i]

  u <- matrix(runif(size * (2 + 5)), nrow = size, ncol = (2 + 5))

  theta.hat.monte.carlo[i] <- mean(g(my.rbeta(size, 2, 5, u)))
}
```

```
theta.hat.antithetic <- c()

for (i in 1:length(sample.sizes)) {
  size <- sample.sizes[i]

  u <- matrix(runif((size / 2) * (2 + 5)), nrow = size / 2, ncol = (2 + 5))

  x <- my.rbeta(size / 2, 2, 5, u)
  x_dash <- my.rbeta(size / 2, 2, 5, 1-u)

  theta.hat.antithetic[i] <- mean(g(c(x, x_dash)))
}
```

```
df3 <- data.frame(sample.size = sample.sizes,
                  theta.hat.mc = theta.hat.monte.carlo,
                  theta.hat.at = theta.hat.antithetic)
```

Table 3: Estimates from Two Different Techniques

sample.size	theta.hat.mc	theta.hat.at
20	0.751372	0.768994
50	0.775342	0.758296
80	0.758218	0.758144
100	0.752671	0.752472

```r
B <- 1000
```

```r
var.monte.carlo <- c()

for (j in 1:length(sample.sizes)) {

  size <- sample.sizes[j]

  theta.hat.monte.carlo <- c()

  for (i in 1:B) {
    u <- matrix(runif(size * (2 + 5)), nrow = size, ncol = (2 + 5))

    theta.hat.monte.carlo[i] <- mean(g(my.rbeta(size, 2, 5, u)))
  }

  var.monte.carlo[j] <- var(theta.hat.monte.carlo)
}
```

```r
var.antithetic <- c()

for (j in 1:length(sample.sizes)) {

  size <- sample.sizes[j]

  theta.hat.antithetic <- c()

  for (i in 1:B) {
    u <- matrix(runif((size / 2) * (2 + 5)), nrow = size / 2, ncol = (2 + 5))

    x <- my.rbeta(size / 2, 2, 5, u)
    x_dash <- my.rbeta(size / 2, 2, 5, 1-u)

    theta.hat.antithetic[i] <- mean(g(c(x, x_dash)))
  }

  var.antithetic[j] <- var(theta.hat.antithetic)
}
```

```r
relative.gain <- ((var.monte.carlo - var.antithetic) / var.monte.carlo) * 100
```

```r
df4 <- data.frame(sample.size = sample.sizes,
                  variance.mc = var.monte.carlo,
                  variance.at = var.antithetic,
                  percentage.reduction = relative.gain)
```

Table 4: Comparison of Variance of Two Techniques

sample.size	variance.mc	variance.at	percentage.reduction
20	0.000703	0.000132	81.251030
50	0.000257	0.000057	77.672460
80	0.000159	0.000037	76.891230
100	0.000127	0.000031	75.728570

✎ We achieve significant reduction in variance of the estimates by using antithetic variable technique.

3

```
g <- function(x) x^2
```

```
theta.hat.monte.carlo <- c()

for (i in 1:length(sample.sizes)) {

  size <- sample.sizes[i]

  u <- matrix(runif(size * (2 + 5)), nrow = size, ncol = (2 + 5))

  theta.hat.monte.carlo[i] <- mean(g(my.rbeta(size, 2, 5, u)))
}
```

```
theta.hat.antithetic <- c()

for (i in 1:length(sample.sizes)) {
  size <- sample.sizes[i]

  u <- matrix(runif((size / 2) * (2 + 5)), nrow = size / 2, ncol = (2 + 5))

  x <- my.rbeta(size / 2, 2, 5, u)
  x_dash <- my.rbeta(size / 2, 2, 5, 1-u)

  theta.hat.antithetic[i] <- mean(g(c(x, x_dash)))
}
```

```
df5 <- data.frame(sample.size = sample.sizes,
                  theta.hat.mc = theta.hat.monte.carlo,
                  theta.hat.at = theta.hat.antithetic)
```

Table 5: Estimates from Two Different Techniques

sample.size	theta.hat.mc	theta.hat.at
20	0.126081	0.096772
50	0.118958	0.105170
80	0.117715	0.112296
100	0.096335	0.109589

True value is

$$
\begin{aligned}
\theta &= E[X^2] \\
&= \int_0^1 x^2 \cdot \frac{1}{\mathcal{B}(2,5)} x(1-x)^4 \, dx \\
&= \frac{1}{\mathcal{B}(2,5)} \int_0^1 x^3 (1-x)^4 \, dx \\
&= \frac{1}{\mathcal{B}(2,5)} \cdot \mathcal{B}(4,5) \\
&= \frac{\Gamma(7)}{\Gamma(2) \cdot \Gamma(5)} \cdot \frac{\Gamma(4) \cdot \Gamma(5)}{\Gamma(9)} \\
&= \frac{6!}{4!} \cdot \frac{3! \cdot 4!}{8!} \\
&= \frac{6}{7 \times 8} \\
&= \frac{3}{28} \\
&= 0.1071429.
\end{aligned}
$$

```
B <- 1000
```

```
var.monte.carlo <- c()

for (j in 1:length(sample.sizes)) {

  size <- sample.sizes[j]

  theta.hat.monte.carlo <- c()

  for (i in 1:B) {
    u <- matrix(runif(size * (2 + 5)), nrow = size, ncol = (2 + 5))

    theta.hat.monte.carlo[i] <- mean(g(my.rbeta(size, 2, 5, u)))
  }

  var.monte.carlo[j] <- var(theta.hat.monte.carlo)
}
```

```
var.antithetic <- c()

for (j in 1:length(sample.sizes)) {

  size <- sample.sizes[j]

  theta.hat.antithetic <- c()
```

```r
  for (i in 1:B) {
    u <- matrix(runif((size / 2) * (2 + 5)), nrow = size / 2, ncol = (2 + 5))

    x <- my.rbeta(size / 2, 2, 5, u)
    x_dash <- my.rbeta(size / 2, 2, 5, 1-u)

    theta.hat.antithetic[i] <- mean(g(c(x, x_dash)))
  }

  var.antithetic[j] <- var(theta.hat.antithetic)
}
```

```r
relative.gain <- ((var.monte.carlo - var.antithetic) / var.monte.carlo) * 100
```

```r
df6 <- data.frame(sample.size = sample.sizes,
                  variance.mc = var.monte.carlo,
                  variance.at = var.antithetic,
                  percentage.reduction = relative.gain)
```

Table 6: Comparison of Variance of Two Techniques

sample.size	variance.mc	variance.at	percentage.reduction
20	0.000617	0.000287	53.495410
50	0.000271	0.000127	53.163210
80	0.000153	0.000070	54.183050
100	0.000125	0.000054	56.463350

✎ We achieve significant reduction in variance of the estimates by using antithetic variable technique.